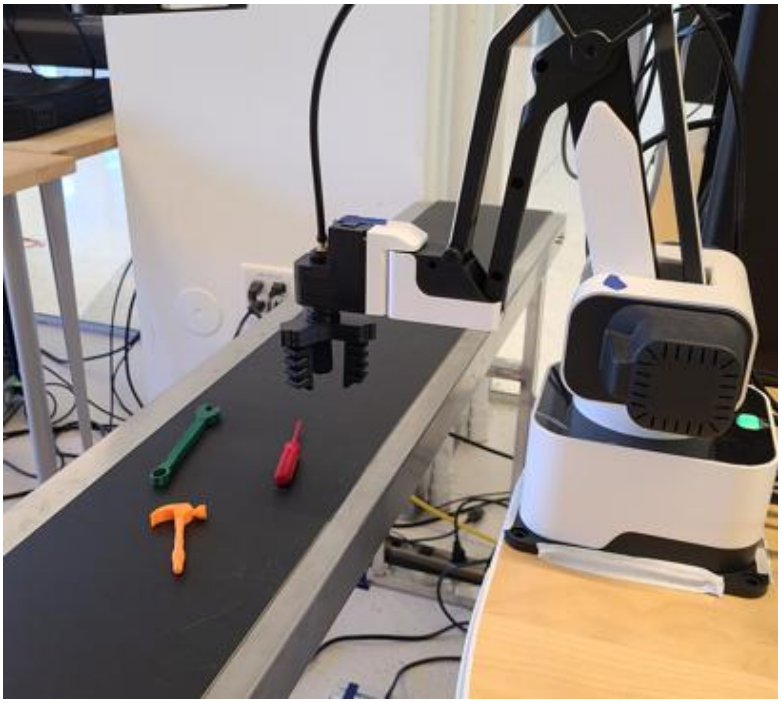
MASTER'S CAPSTONE / SANTA CLARA UNIVERSITY / 2025

AI-Enabled Multi-Task Robot Workcell

Stereo perception. Tool characterization. Robotic manipulation.



Experimental workcell with a ZED stereo camera, conveyor, and Rotrics DexArm.
Photograph reproduced from the author's capstone report.

THE PROJECT

Developed a vision-guided workcell to identify hammers, wrenches, and screwdrivers and execute robotic pick-and-place operations. The system combines YOLOv8 detection, stereo depth, tool characterization, calibrated robot targets, and visual pickup verification.

MY ROLE

Developed all submitted code and the work described in the report: perception, calibration, socket communication, DexArm integration, retry logic, system testing, and technical documentation.

Python | YOLOv8 | OpenCV | ZED SDK | TCP/JSON | DexArm

dakshithaama1994@gmail.com | LinkedIn profile

01 / ENGINEERING APPROACH

From camera data to robot action

The challenge

Picking requires more than an object label: the arm needs a reachable position, suitable gripper alignment, and a response to unsuccessful grasps. This project integrates those steps in a modular Python implementation.

PERCEPTION SERVER	MONITOR CLIENT	ROBOT CLIENT
ZED + YOLOv8	**Tool attributes**	**DexArm control**

TCP sockets + newline-delimited JSON; separate verification request channel.

Perception and characterization

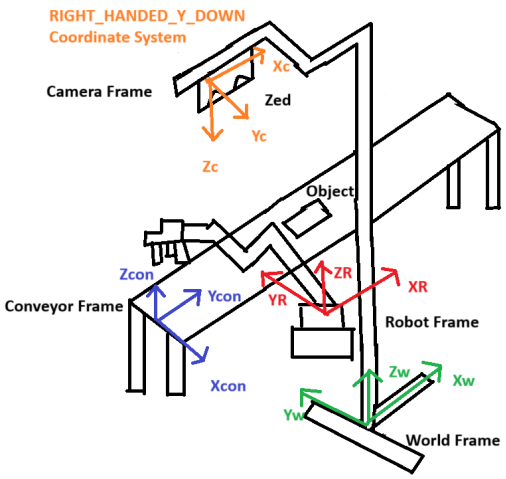
YOLOv8 detects tool classes. Stereo depth supports 3D position and dimension estimates; image processing estimates color and orientation. The software also calculates motion information.

Coordinate mapping and manipulation

Calibrated mappings convert camera measurements to robot targets. The robot client checks workspace bounds, aligns the end-effector, and executes grasp, transfer, placement, and return motions.

Command handling and recovery

Separate listener and control threads use a queue and busy flag to coordinate commands. A visual check looks for a matching tool near its previous position; failed checks trigger bounded retries.



Coordinate frames

Camera, conveyor, robot, and world frames establish spatial references for the workcell.

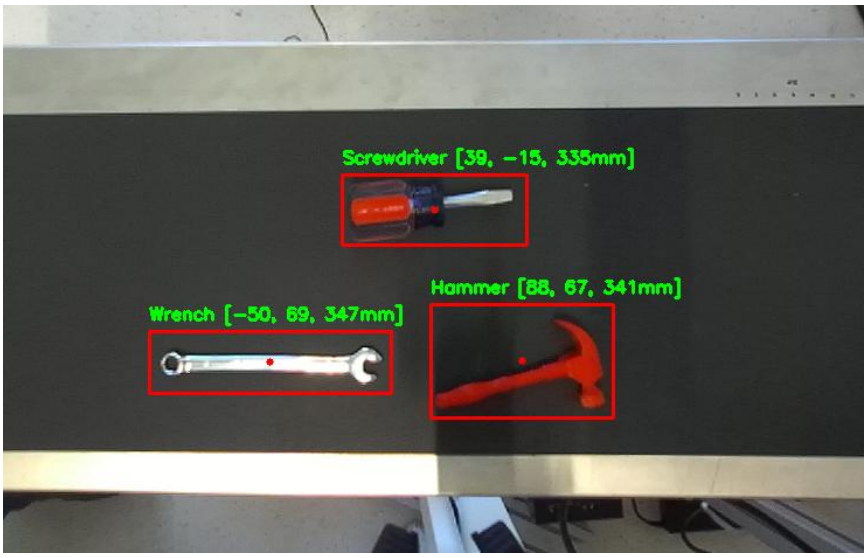
Implementation boundary

The submitted retry loop reuses the received target position and angle. Visual verification infers pickup from absence near the original location; it does not directly sense a secure grasp.

02 / VISION IN ACTION

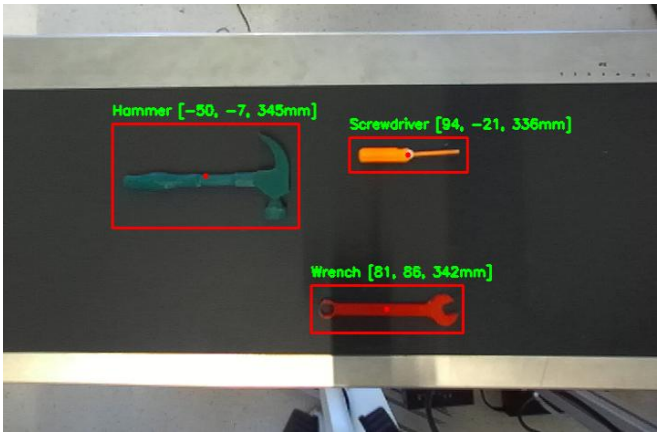
Tool detection and color recognition

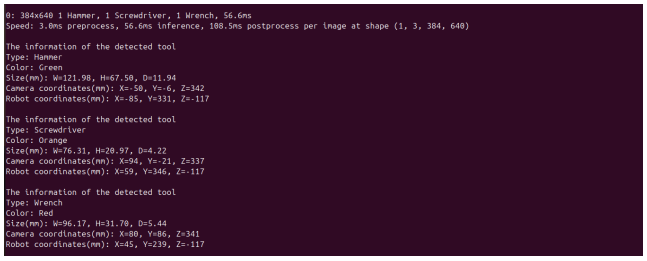
Object detection + 3D position



YOLOv8 identifies a screwdriver, wrench, and hammer. Labels show camera-frame coordinates in millimeters. Source: report Figure 3.3.

Different tools, different colors





Left: scene containing tools of different colors. Right: console output reports class, color, dimensions, and camera/robot coordinates. Source: report Figure 3.4.

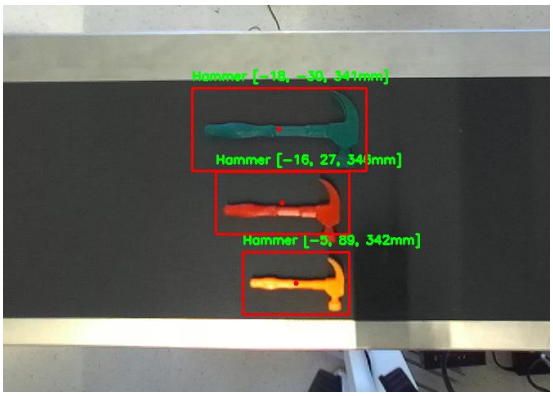
Color processing

The report describes HSV color masks and morphological processing. Lighting, shadows, and background color remain important limitations.

03 / DEPTH-BASED CHARACTERIZATION

Size and color characterization

Same tool class, three sizes




```
The tool details are:
[Client] Received: {'Type': 'Hammer', 'Color': 'Green', 'Size(mm)': [125.03, 54.33, 16.76], 'Size Category': 'Large'}

The tool details are:
[Client] Received: {'Type': 'Hammer', 'Color': 'Orange', 'Size(mm)': [75.71, 45.0, 11.51], 'Size Category': 'Small'}

The tool details are:
[Client] Received: {'Type': 'Hammer', 'Color': 'Red', 'Size(mm)': [96.94, 44.66, 16.37], 'Size Category': 'Medium'}

The tool details are:
[Client] Received: {'Type': 'Hammer', 'Color': 'Green', 'Size(mm)': [125.16, 57.94, 15.53], 'Size Category': 'Large'}

The tool details are:
[Client] Received: {'Type': 'Hammer', 'Color': 'Orange', 'Size(mm)': [75.65, 44.9, 11.25], 'Size Category': 'Small'}

The tool details are:
[Client] Received: {'Type': 'Hammer', 'Color': 'Red', 'Size(mm)': [94.79, 44.62, 16.65], 'Size Category': 'Medium'}

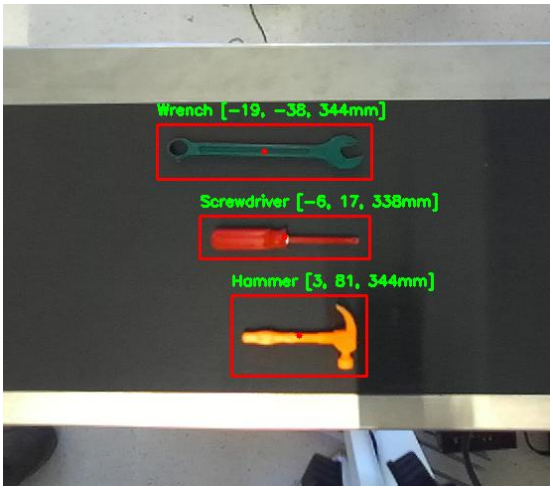
The tool details are:
[Client] Received: {'Type': 'Hammer', 'Color': 'Green', 'Size(mm)': [125.34, 57.31, 16.17], 'Size Category': 'Large'}

The tool details are:
[Client] Received: {'Type': 'Hammer', 'Color': 'Orange', 'Size(mm)': [75.68, 44.87, 11.48], 'Size Category': 'Small'}

The tool details are:
[Client] Received: {'Type': 'Hammer', 'Color': 'Red', 'Size(mm)': [96.8, 42.86, 15.98], 'Size Category': 'Medium'}
```

Hammers are categorized as Small, Medium, and Large. The console shows estimated dimensions in millimeters and color labels. Source: report Figure 3.6.

Different tool classes and attributes




```
The tool details are:
[Client] Received: {'Type': 'Hammer', 'Color': 'Orange', 'Size(mm)': [79.29, 47.34, 12.82], 'Size Category': 'Small'}

The tool details are:
[Client] Received: {'Type': 'Wrench', 'Color': 'Green', 'Size(mm)': [125.71, 33.69, 10.43], 'Size Category': 'Large'}

The tool details are:
[Client] Received: {'Type': 'Screwdriver', 'Color': 'Red', 'Size(mm)': [100.49, 26.9, 8.69], 'Size Category': 'Medium'}

The tool details are:
[Client] Received: {'Type': 'Hammer', 'Color': 'Orange', 'Size(mm)': [79.25, 47.39, 12.63], 'Size Category': 'Small'}

The tool details are:
[Client] Received: {'Type': 'Wrench', 'Color': 'Green', 'Size(mm)': [125.65, 33.22, 10.37], 'Size Category': 'Large'}

The tool details are:
[Client] Received: {'Type': 'Screwdriver', 'Color': 'Red', 'Size(mm)': [102.27, 26.93, 7.7], 'Size Category': 'Medium'}

The tool details are:
[Client] Received: {'Type': 'Hammer', 'Color': 'Orange', 'Size(mm)': [79.28, 47.44, 12.89], 'Size Category': 'Small'}

The tool details are:
[Client] Received: {'Type': 'Wrench', 'Color': 'Green', 'Size(mm)': [125.59, 34.31, 10.37], 'Size Category': 'Large'}

The tool details are:
[Client] Received: {'Type': 'Screwdriver', 'Color': 'Red', 'Size(mm)': [102.37, 26.91, 8.08], 'Size Category': 'Medium'}

The tool details are:
[Client] Received: {'Type': 'Hammer', 'Color': 'Orange', 'Size(mm)': [79.31, 47.19, 13.05], 'Size Category': 'Small'}

The tool details are:
[Client] Received: {'Type': 'Wrench', 'Color': 'Green', 'Size(mm)': [125.57, 34.31, 10.45], 'Size Category': 'Large'}
```

Hammer, screwdriver, and wrench examples paired with transmitted tool attributes. Source: report Figure 3.7.

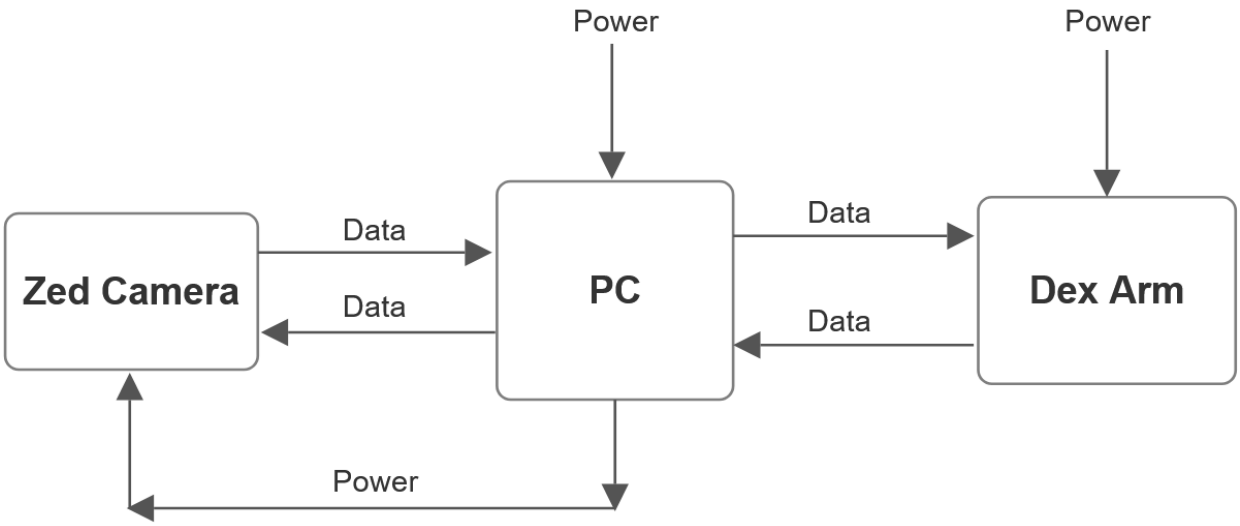
Method

Sample stereo 3D points inside each detection, reject invalid values and percentile outliers, estimate axis-aligned dimensions, and apply calibrated size thresholds. These are depth-derived estimates rather than metrology-certified measurements.

04 / INTEGRATION DESIGN

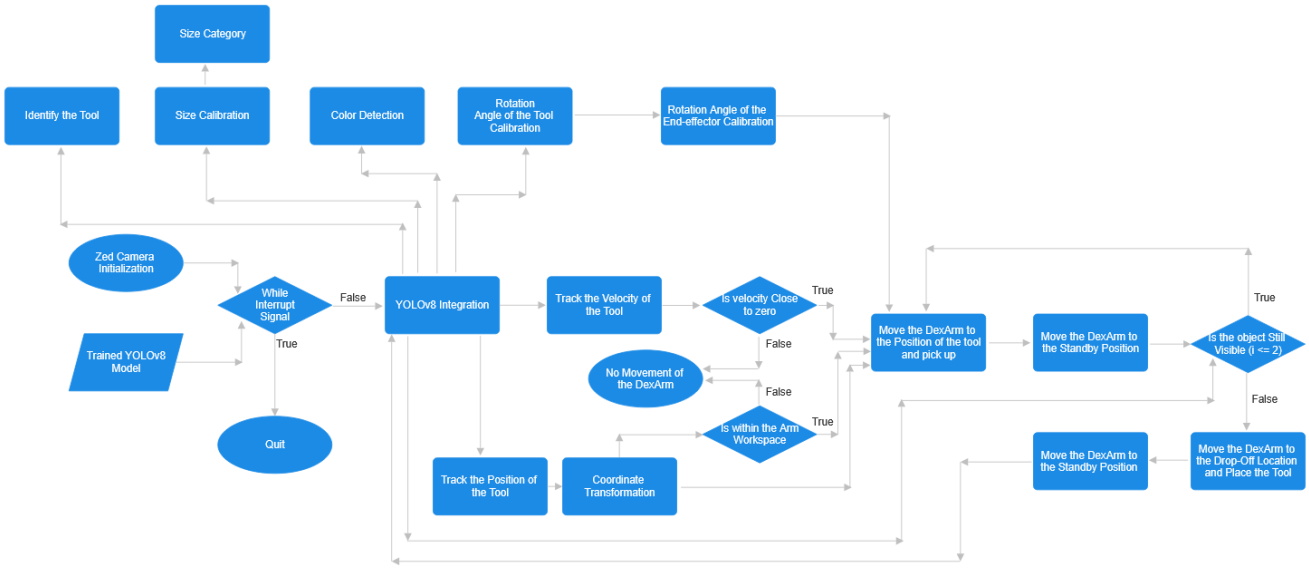
Hardware and high-level software

Component block diagram



Author-supplied diagram of workcell components and interfaces.

High-level software flowchart



Design-level workflow from the report. For submitted-code behavior and retry limitations, see page 2; the diagram is not a claim that every design step is enforced in the final scripts.

05 / TESTING & REFLECTION

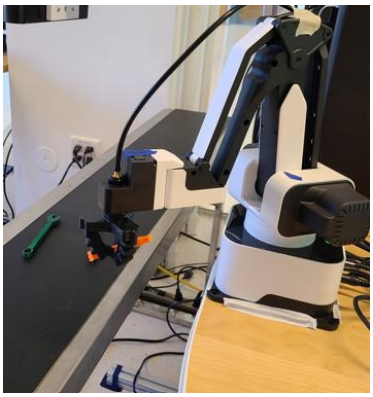
Measured performance and lessons

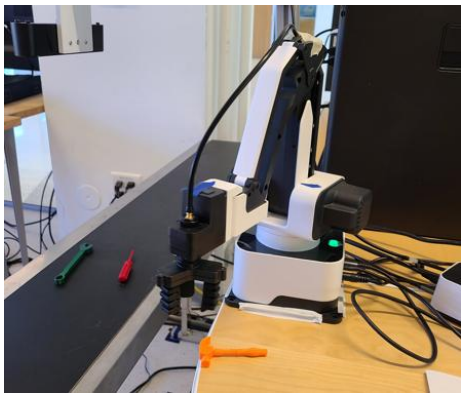
Reported mean pickup success ratio

10 trials per configuration; three tools per trial. Metric: the mean of per-trial successful-pickup-to-attempt ratios, not detection accuracy.



*Test 3 is described in the report as orientation alignment with position/orientation feedback updates. The submitted code reuses its received target during retries, so those updates are not established by this code snapshot. Values above are report-recorded experimental results.





Pickup and placement sequences documented in the capstone report.

Engineering lessons and next steps

Lighting and dark backgrounds affected visual processing. Soft-gripper stability and payload constraints influenced handling, and lightweight 3D-printed tools were used. Future work: refresh targets before retries, add stronger grasp confirmation, improve lighting robustness, and measure cycle time and localization error.

Evidence: Capstone Design Project Report, Summer 2025, pp. 38-51; submitted sender4.py, s1client4.py, and s2client4.py. Contribution attribution confirmed by the author. Portfolio prepared from supplied material; no new hardware tests performed.